

Executive Overview

SatQuery AI bridges the gap between complex Earth Observation (EO) data and non-expert operational users. By combining multi-sensor satellite imagery (Optical/Multispectral and Synthetic Aperture Radar) with an *agentic orchestration framework*, SatQuery AI replaces fragmented GIS tools with natural-language query resolution. The platform automatically interprets intent, routes tasks to specialized models, performs cross-modal and multi-temporal reasoning, and produces auditable, evidence-grounded outputs.

1. CORE OPERATIONAL CHALLENGES & INNOVATIONS

● Current Technical Bottlenecks

- **Fragmented Workflows:** Conventional AI tools solve isolated tasks (e.g., land-cover classification OR change detection), forcing users to manually stitch outputs together.
- **High Domain Barrier:** Users must understand satellite orbits, sensor geometries, band configurations, and complex GIS pipelines.
- **Single-Image Limitations:** Optical data fails in cloud cover or night; SAR lacks fine spectral context. Single temporal frames miss dynamic shifts.

● SatQuery AI Innovations

- **Agentic Orchestration:** Dynamic query decomposition and model routing based on sensor modality and task parameters.
- **Cross-Modal Synergies:** Fuses co-registered Optical (spectral details) and SAR (structural/all-weather) data for unified analysis.
- **Bi-Temporal Reasoning:** Spatiotemporal change detection and change-based Visual Question Answering (CDVQA).

2. MULTI-MODAL INPUT SCOPE

The system natively supports professional geospatial image formats alongside standard reference inputs:

Single Image

Optical / Multispectral or SAR

Supports VQA, scene description/captioning, and text-guided spatial region grounding.

Cross-Modal Pair

Co-registered Optical + SAR

Joint feature extraction combining spectral land cover with structural SAR backscatter.

Bi-Temporal Pair

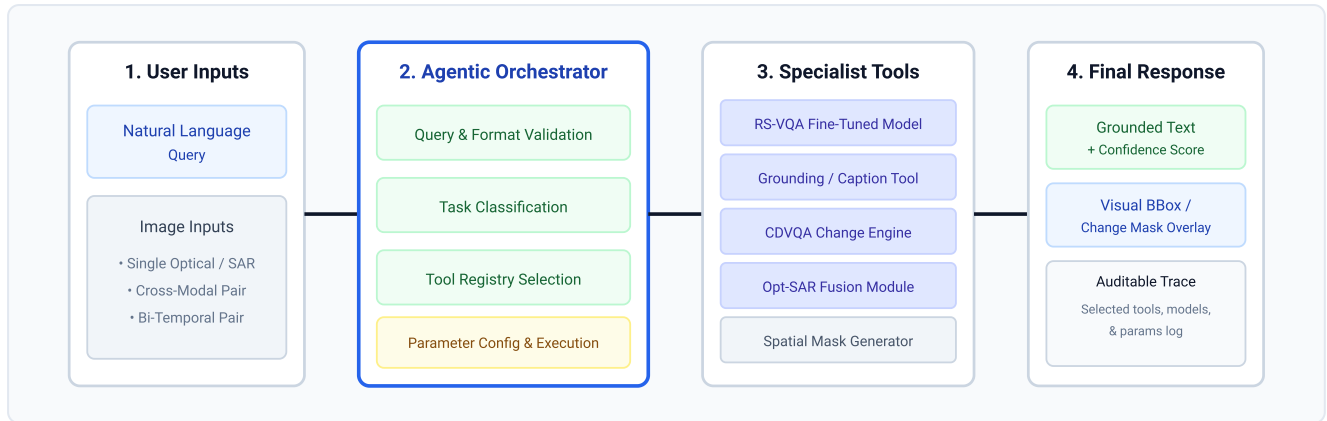
Time T₁ + Time T₂ Images

Spatiotemporal change understanding, change captioning, and localized mask generation.

Supported File Formats: Primary operational format: GeoTIFF / TIFF (retaining coordinate reference systems and metadata). PNG / JPEG accepted strictly for evaluation against public benchmark datasets.

3. SYSTEM ARCHITECTURE & AGENTIC WORKFLOW

SatQuery AI uses an autonomous controller that processes user input through a transparent, multi-stage pipeline, executing specialist models from a predefined model registry.



4. CORE FUNCTIONAL MODULES & BENCHMARKS

Functional Scope	Mandatory Task	Primary Dataset / Benchmark	Key Capabilities & Outputs
Domain Adaptation	Fine-Tuned VLM Component	BigEarthNet.txt / Open-Source RS	Adapts generic vision-language features to satellite spectral characteristics & terminology.
Single-Image Baseline	Visual Question Answering + Grounding / Captioning	VRSBench RSVQA	Natural language query parsing, land-cover identification, object grounding with visual bounding boxes.
Bi-Temporal Change	Change Description / Change-VQA	CDVQA	Detects urban expansion, deforestation, disaster damage between T_1 and T_2 ; outputs change maps.
Cross-Modal Analysis	Joint Optical-SAR Fusion	ISRO / SAC Cartosat-2S & RISAT Pair	Extracts complementary structural (SAR) and spectral (optical) details for all-weather monitoring.

5. REPRESENTATIVE QUERY SCENARIOS

Single Image Analysis

"Describe the land-cover and major objects visible in this image."

"Highlight the water body referred to in the query."

Multi-Image Change & Cross-Modal

"What changed between these two dates, and where did the change occur?"

"Use optical and SAR images together to identify built-up and water regions."

6. EVALUATION & JUDGING CRITERIA MATRIX

Solutions will be benchmarked on public datasets and an exclusive **ISRO/SAC evaluation dataset** containing pre-georeferenced Cartosat-2S (optical) and RISAT (SAR) imagery. Metric scores are normalized before aggregation.

Evaluation Metric	Target Scope	Benchmark Source	Evaluation Criteria & Primary Focus
VQA Accuracy & BLEU/ROUGE	Single-Image VQA & Captioning	VRSBench / RSVQA	Measures accuracy of answers and linguistic similarity of generated scene descriptions.
mIoU / Bounding Box mAP	Text-Guided Region Grounding	VRSBench	Precision of spatial localization for targeted geographic features and objects.
Change-VQA Score & F1-Score	Bi-Temporal Change Analysis	CDVQA / ISRO Evaluation Set	Correctness of change identification and spatial alignment of change detection masks.
Cross-Modal F1-Score	Joint Optical-SAR Extraction	ISRO Cartosat-2S & RISAT Pairs	Feature extraction quality leveraging complementary SAR backscatter and optical spectra.
Agent Orchestration Precision	Agentic Workflow	Execution Trace Evaluation	Accuracy of tool selection, proper parameter passing, and validity of input compatibility checks.

7. AGENTIC EXECUTION SUMMARY & AUDITABILITY

Internal reasoning thoughts are hidden from evaluation; only the **observable execution trace** is evaluated. SatQuery AI outputs an auditable summary alongside every query answer:

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[EXECUTION TRACE SUMMARY]
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• Detected Task Type : Bi-Temporal Change Visual Question Answering (CDVQA)
• Input Validation : T1 (GeoTIFF, Optical, 0.5m CRS: EPSG:32643) PASSED
T2 (GeoTIFF, Optical, 0.5m CRS: EPSG:32643) PASSED
• Selected Workflow : [Tool: GeoAlign] -> [Model: RS-ChangeFormer] -> [Model: CDVQA-LLM]
• Executed Parameters : Threshold=0.45, Mask_Format=GeoJSON, Resolution=0.5m
• Confidence Score : 94.2%
• Visual Artifacts : Change_Mask_Overlay.png, Grounded_Bounding_Box.json
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8. EXPECTED DELIVERABLES & IMPLEMENTATION SCOPE

Interactive Web Application (GUI) • Drag-and-drop support for single GeoTIFFs, cross-modal pairs, and bi-temporal pairs. • Interactive geospatial visualizer with vector bounding boxes and change mask overlays. • Side-by-side execution trace viewer with confidence metrics. • One-click exportable PDF analytical reports containing grounded evidence.	Codebase & Model Pipeline • Domain-adapted vision-language model trained/fine-tuned on BigEarthNet.txt data. • Predefined Model Registry featuring VQA, Grounding, Change Engine, and Optical-SAR Fusion. • Agentic Controller Backend with automated input compatibility & error handling. • Comprehensive test suite and demonstration scripts for ISRO/SAC evaluation datasets.
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